

Mathematical Analysis of Jamming and Simulation Environment

Communication Jamming

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- **Likelihood Function** $f(y|s)$

Represents how plausible a transmitted signal s is, given a received signal y .

- **Characteristic Function**

Transforms a linear sum of variables into a product of exponentials, simplifying statistical analysis.

- **Q-function** $Q(x)$

The integral of the standard normal distribution from x to ∞ .

- **BPSK**

A digital modulation scheme: recognizes 1 as amplitude A and 0 as $-A$.

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Maximum Likelihood Detection (MLD)

Assuming all symbols $s \in S$ have equal prior probabilities, we maximize the likelihood function $f(y|s)$:

$$\hat{s} = \arg \min_{s \in S} |y - s|^2$$

This reduces to minimizing the Euclidean distance in the signal space. Although expressed mathematically, it is essentially a simple decision of which symbol, 1 or 0, is closer to the received value.

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The system includes natural AWGN (w) and jamming noise (z). The received signal y is defined as:

$$y = s + w + z$$

Using the characteristic function, the total noise can be modeled as a combined Gaussian distribution: $\mathcal{N}(0, \sigma_w^2 + \sigma_z^2)$.

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Information Loss (1)

Analyzing the limit as jamming power approaches infinity ($\sigma_z^2 \rightarrow \infty$):

$$\begin{aligned}\lim_{\sigma_z^2 \rightarrow \infty} P_e &= \lim_{\sigma_z^2 \rightarrow \infty} Q\left(\frac{A}{\sqrt{\sigma_w^2 + \sigma_z^2}}\right) \\ &= Q(0) = \frac{1}{2}\end{aligned}$$

The bit error rate converges to 0.5, indicating a total loss of information.

Information Loss (2)

From the Shannon-Hartley theorem:

$$\begin{aligned} C &= \lim_{\sigma_z^2 \rightarrow \infty} B \log_2 \left(1 + \frac{A^2}{\sigma_w^2 + \sigma_z^2} \right) \\ &= B \log_2(1) = 0 \end{aligned}$$

The channel capacity vanishes.

Visualizing Jamming Effect

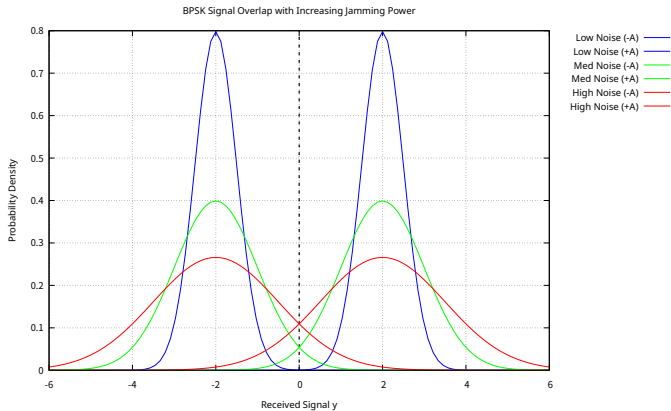


Figure: BPSK Signal Overlap under Jamming

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Jamming Power Dependency

The analysis shows that the communication reliability is heavily dependent on the jamming noise intensity. Higher jamming power severely degrades communication quality, ultimately leading to a random guess state.

Environment Setup

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Deployed Pyjama on the shikharlab-01 server.

- Directory: `jamming_env` (under home)
- Files: `Dockerfile`, `docker-compose.yml`
- Access: SSH to server, then Terminal or VS Code for container access.
- Recommendation: VS Code (best for Jupyter environment).
- Access: Port 8080 exposed on the server.
- Note: Performance may differ due to the remote environment.

Future Plan

Future Plan

- Study Reinforcement Learning (RL) in Dr. Wei's seminar.
- Study Game Theory to apply to anti-jamming strategies and prediction.
- Develop a Python program for basic jamming simulation.

References

References I

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